

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

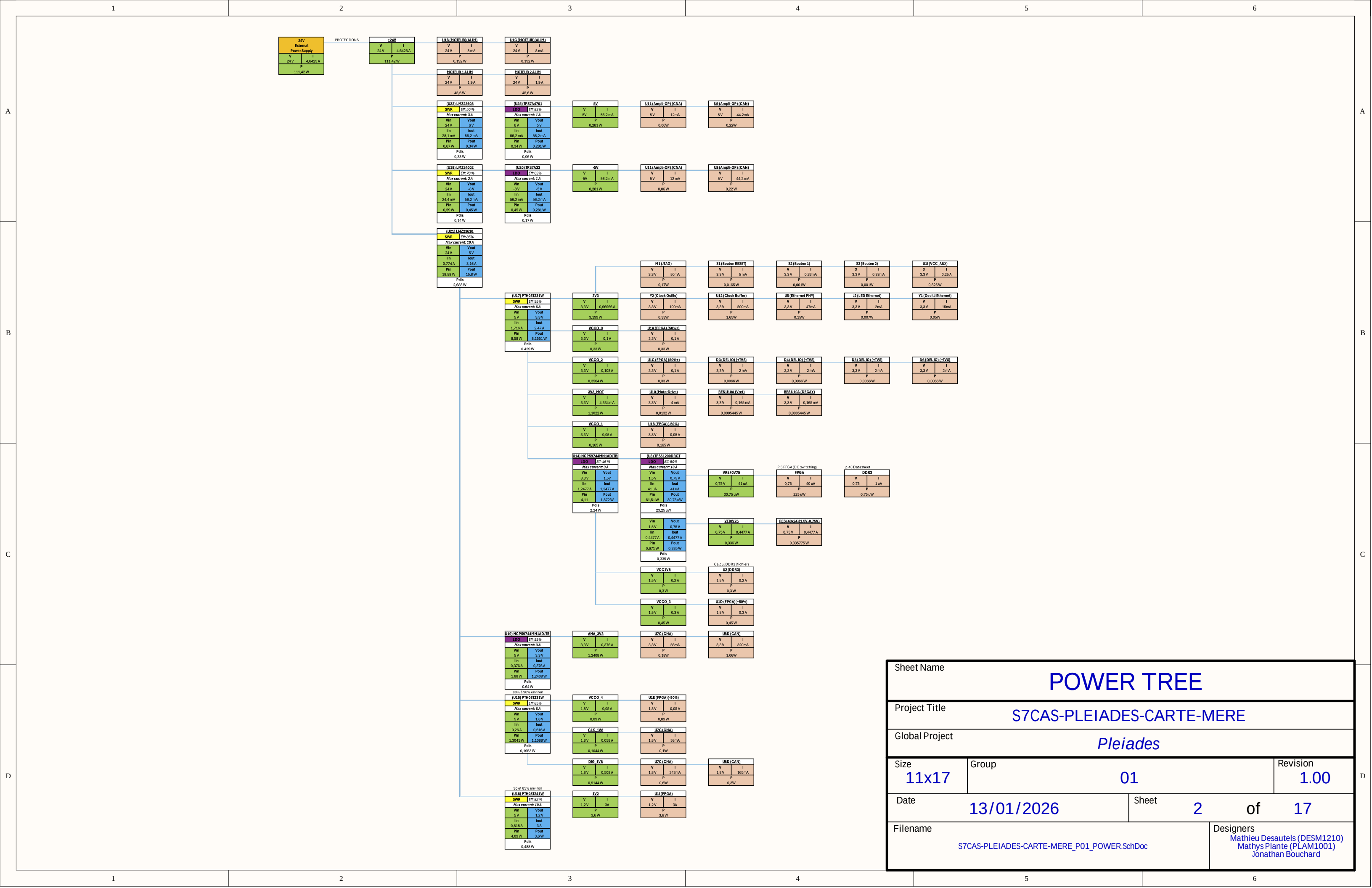
Date : 13/01/2026

Revision history	

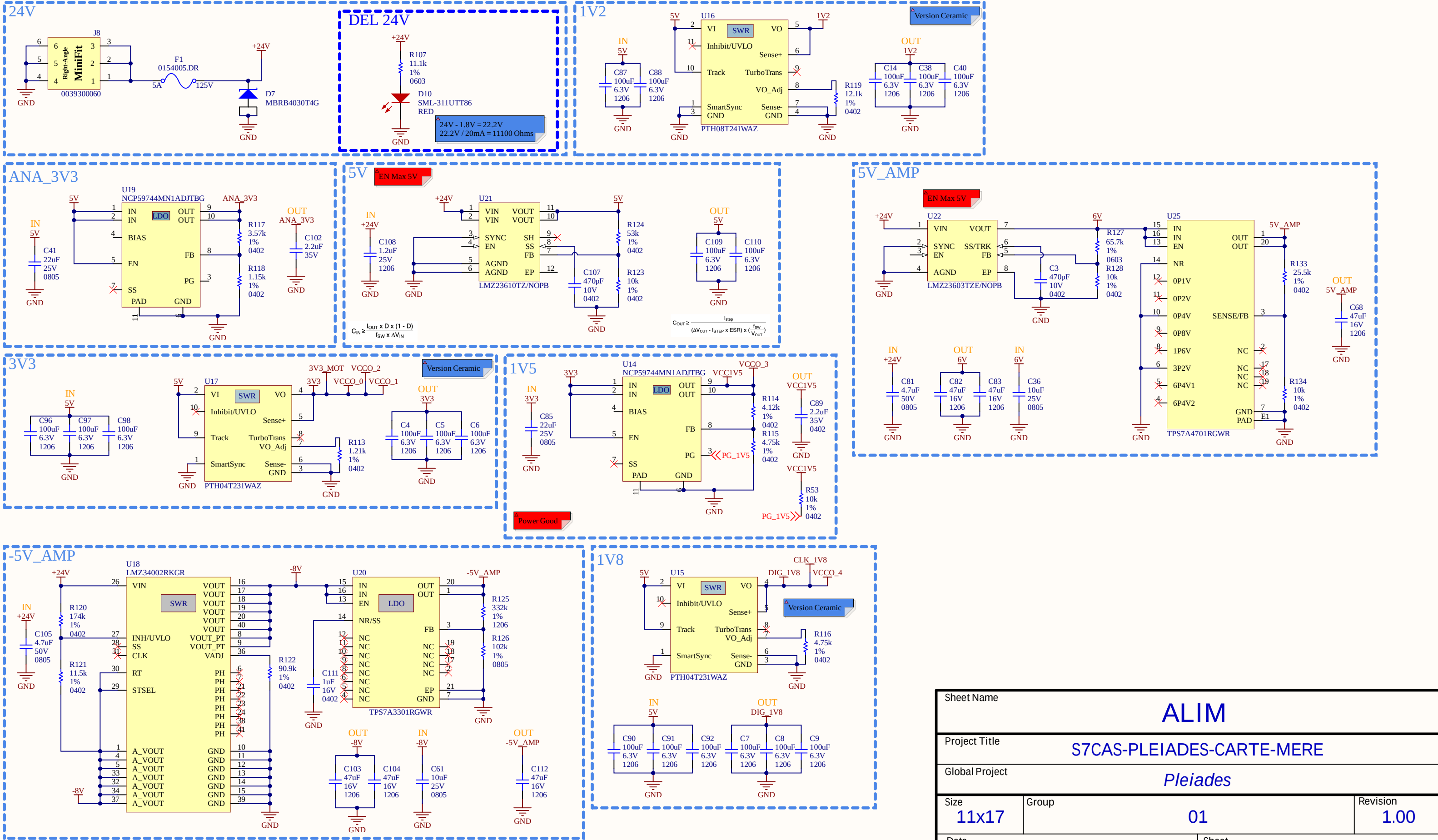
Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
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Project Title		
S7CAS-PLEIADES-CARTE-MERE		
Global Project		
Pleiades		
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S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

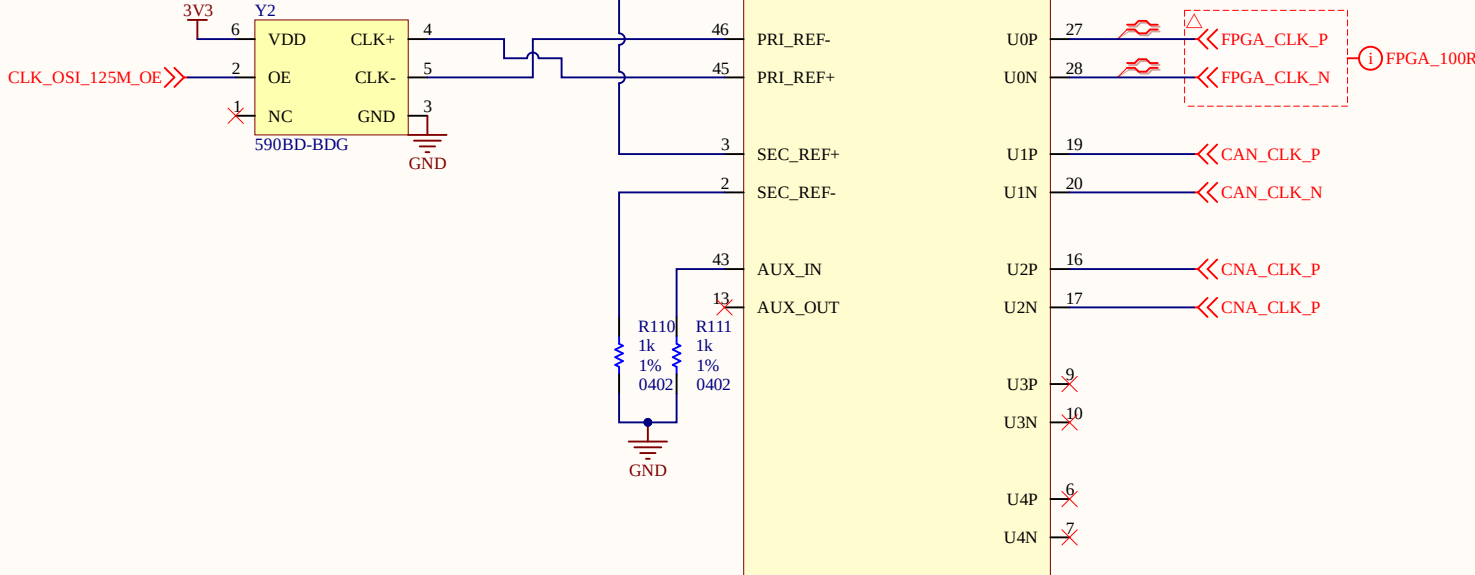


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Global Project		Pleiades	
Size	Group	Revision	
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Date	13/01/2026	Sheet	2 of 17
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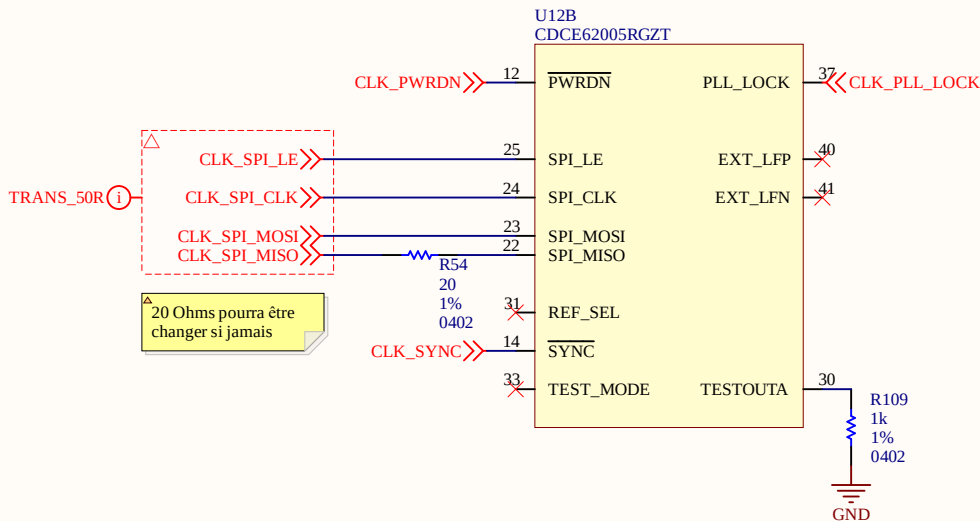
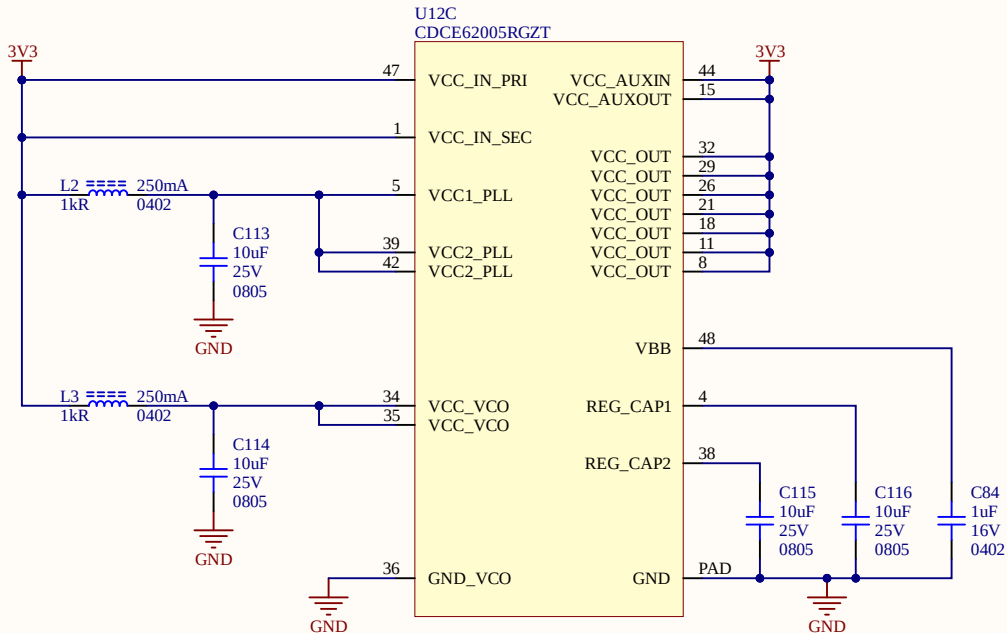


Sheet Name			ALIM		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
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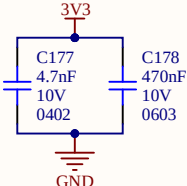
CLOCK



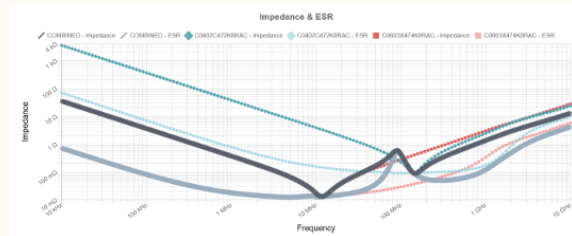
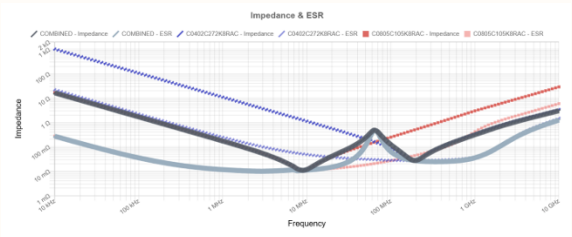
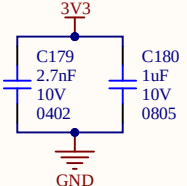
CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf
CLOCK BUFFER - CDCE62005.pdf



DECOUPLAGE CLOCK



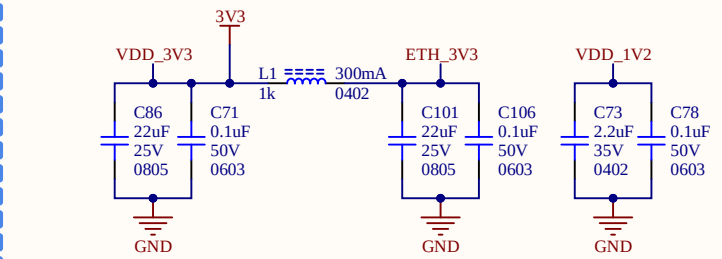
DECOUPLAGE BUFFER



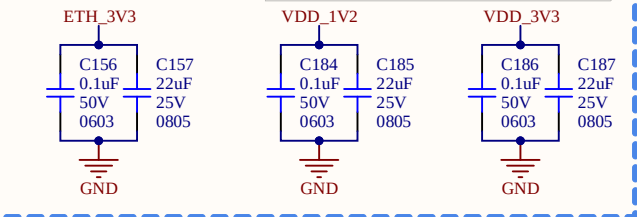
-Quand tu es en LVDS faut que branche le SEC à une résistance (CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf)
-CAN et CNA pas fait pour la pair différentiel puisqu'elle est directement dans le CAN et CNA

Sheet Name			CLOCK		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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11x17	01			1.00	
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S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

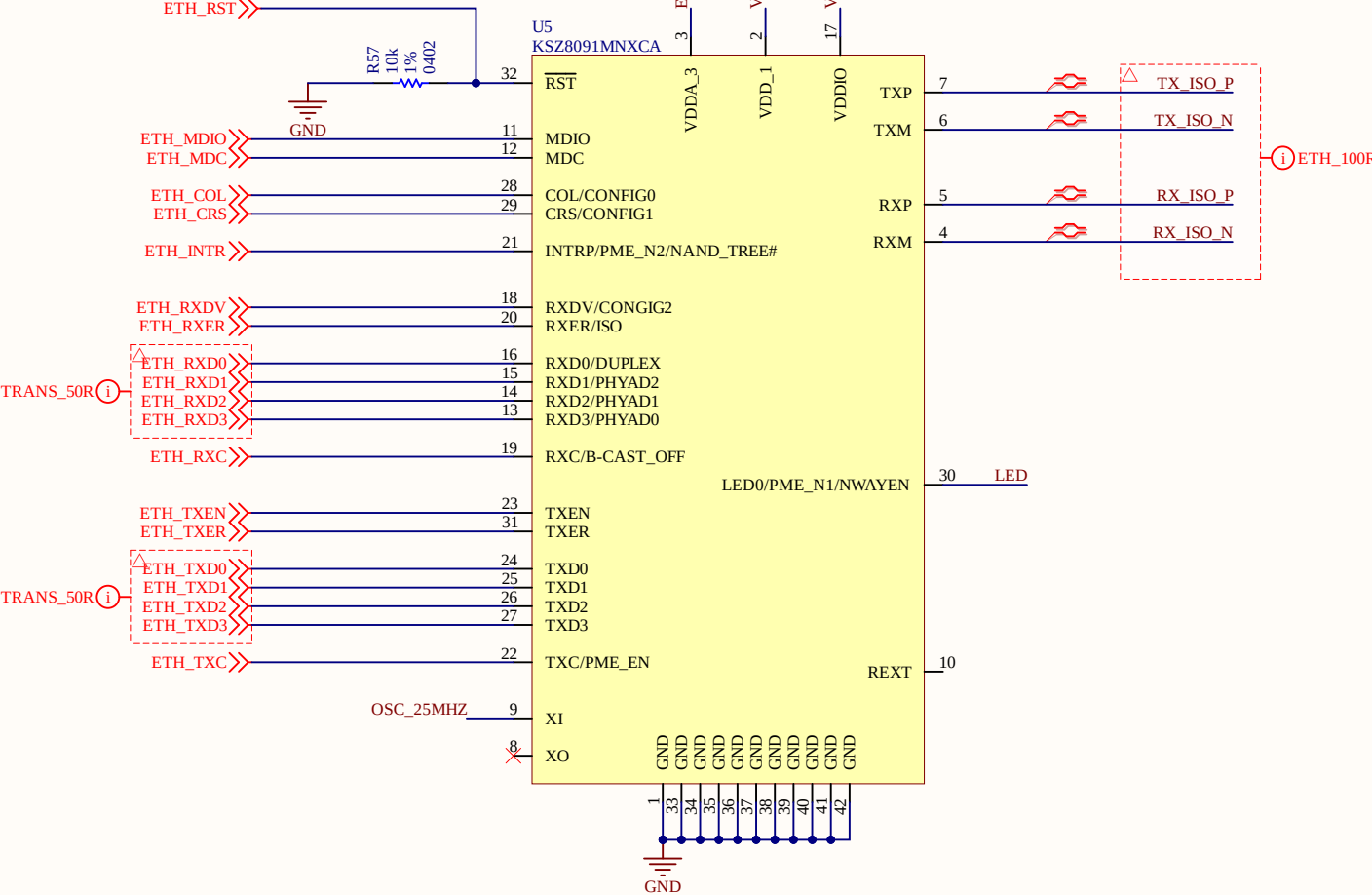
ETHERNET
ALIMENTATION



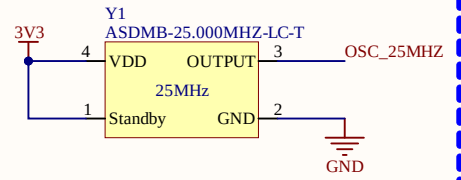
DECOUPLAGE



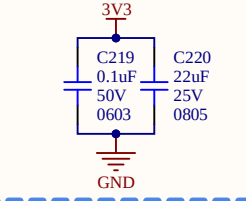
ETHERNET - PHY - KSZ8091MNXCA.pdf



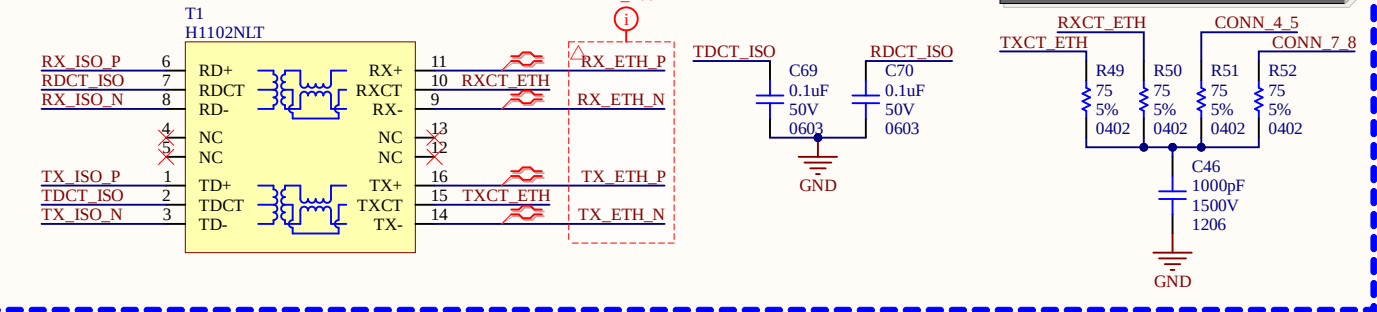
OSCILLI



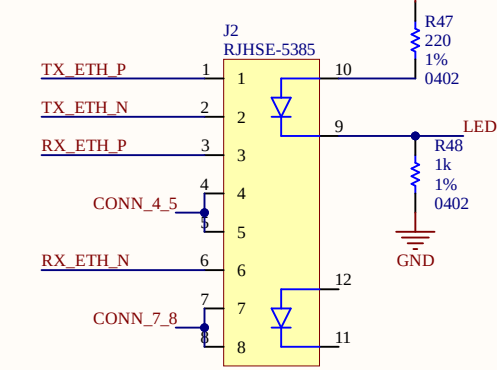
DECOUPLAGE



ISO

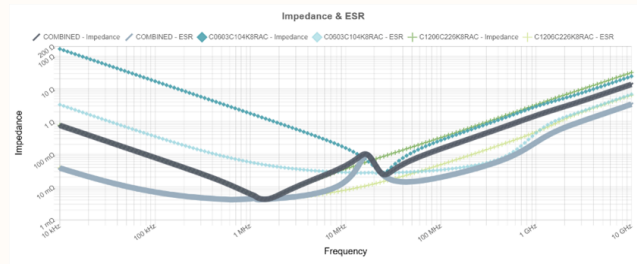
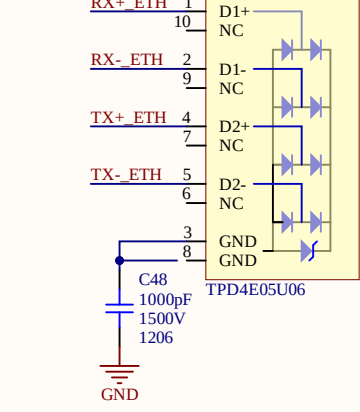


CONNECTEUR



In #	Wire Color Legend	Signal
1	White/Green	TX+
2	Green	TX-
3	White/Orange	RX+
4	Blue	TRD2+
5	White/Blue	TRD2-
6	Orange	RX-
7	White/Brown	TRD3+
8	Brown	TRD3-

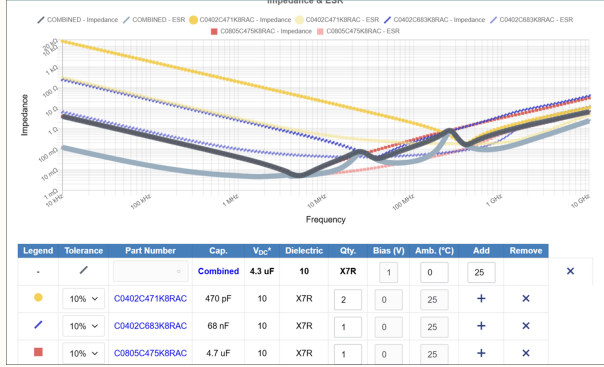
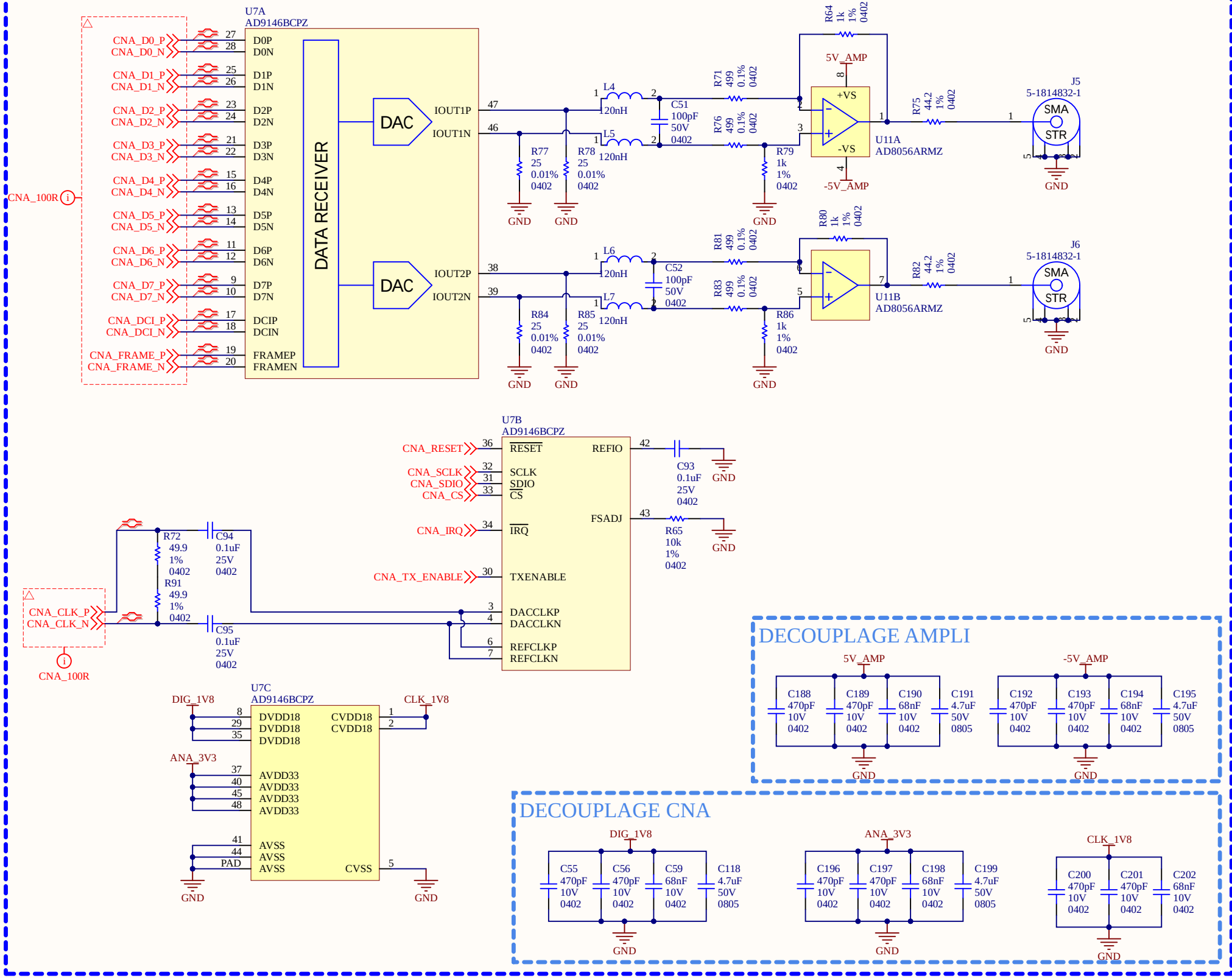
ESD



-Isolation du ETHERNET 1.5kV
-Ajout d'un condensateur de 1.5kV
-Ligne de transmission de 50R
-TX et RX voir le référence design pourquoi c'est ligne de transmission sans différentiel

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S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CNA (DAC)



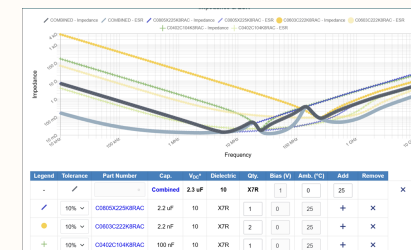
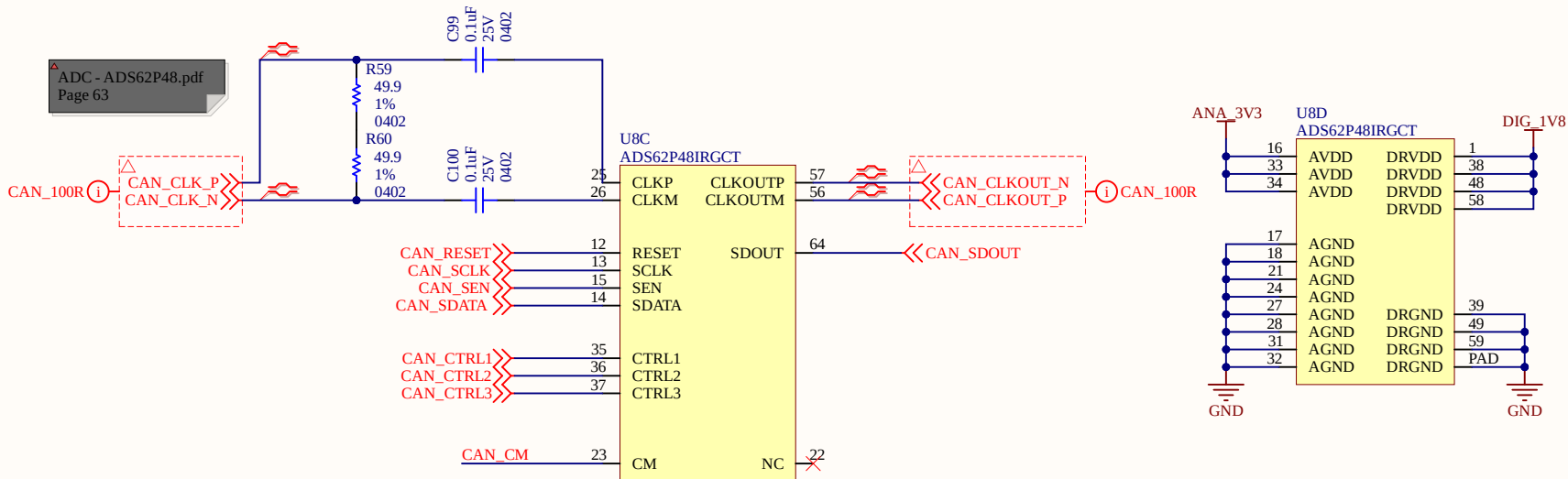
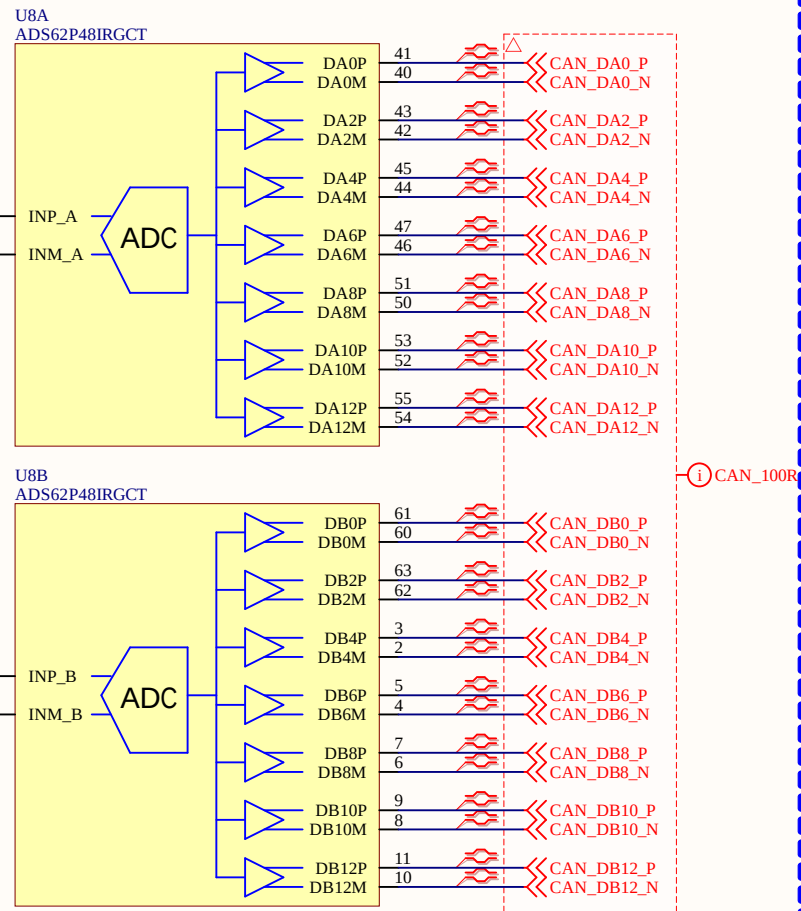
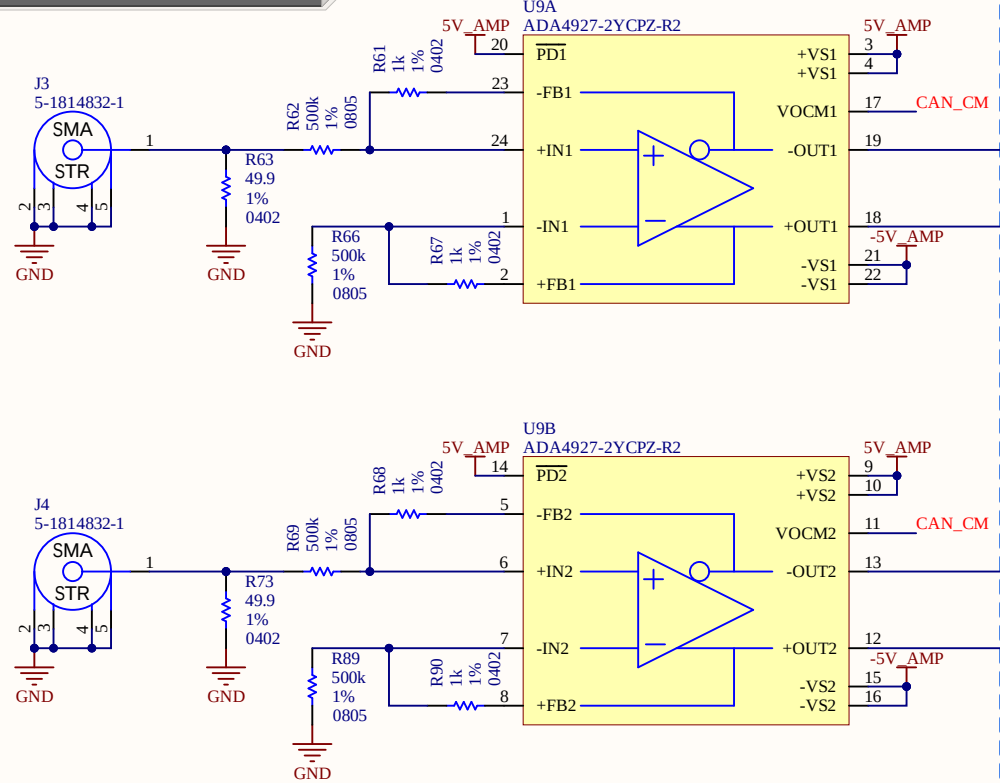
-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Ampli-op aimerait un gain de 2 avec 1V à la sortie, il faut donc 0.5V en entree
-DAC sortie au maximum c'est 20mA, donc avec le 25ohms c'est 0.5V
-On réussi à avoir 1V à la sortie
-5ohms à la sortie à 40MHz donc 45ohms pour adapter
-Filtre passe bas de deuxième ordre (0 à 40MHz), point de coupure 45MHz = L et C
-Receiver Differential impedance 80 to 120ohms

Sheet Name		CNA	
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S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

CAN (ADC)

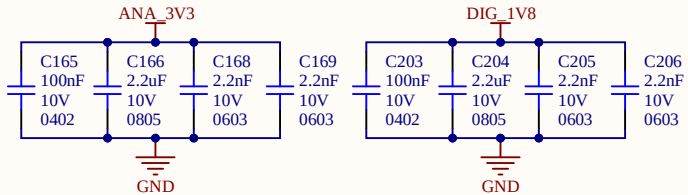
DECOUPLAGE ANALOG

OPAMP - ADA4927-1_ADA4927-2.pdf
P.20

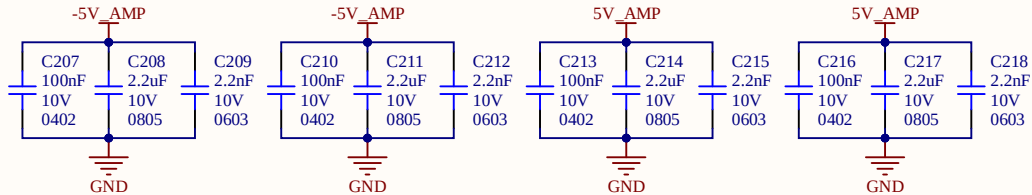


-Choix de OP-AMP pour isolé et pour passer d'analog a digital
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Nous avons pris uniquement AMPLI-OP puisque transfo bloque le DC et on a une composante DC

DECOUPLAGE CAN

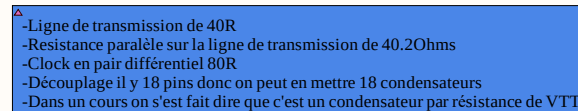
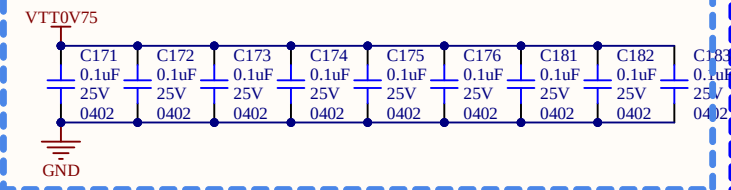


DECOUPLAGE AMPLI



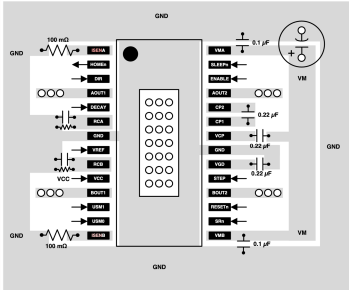
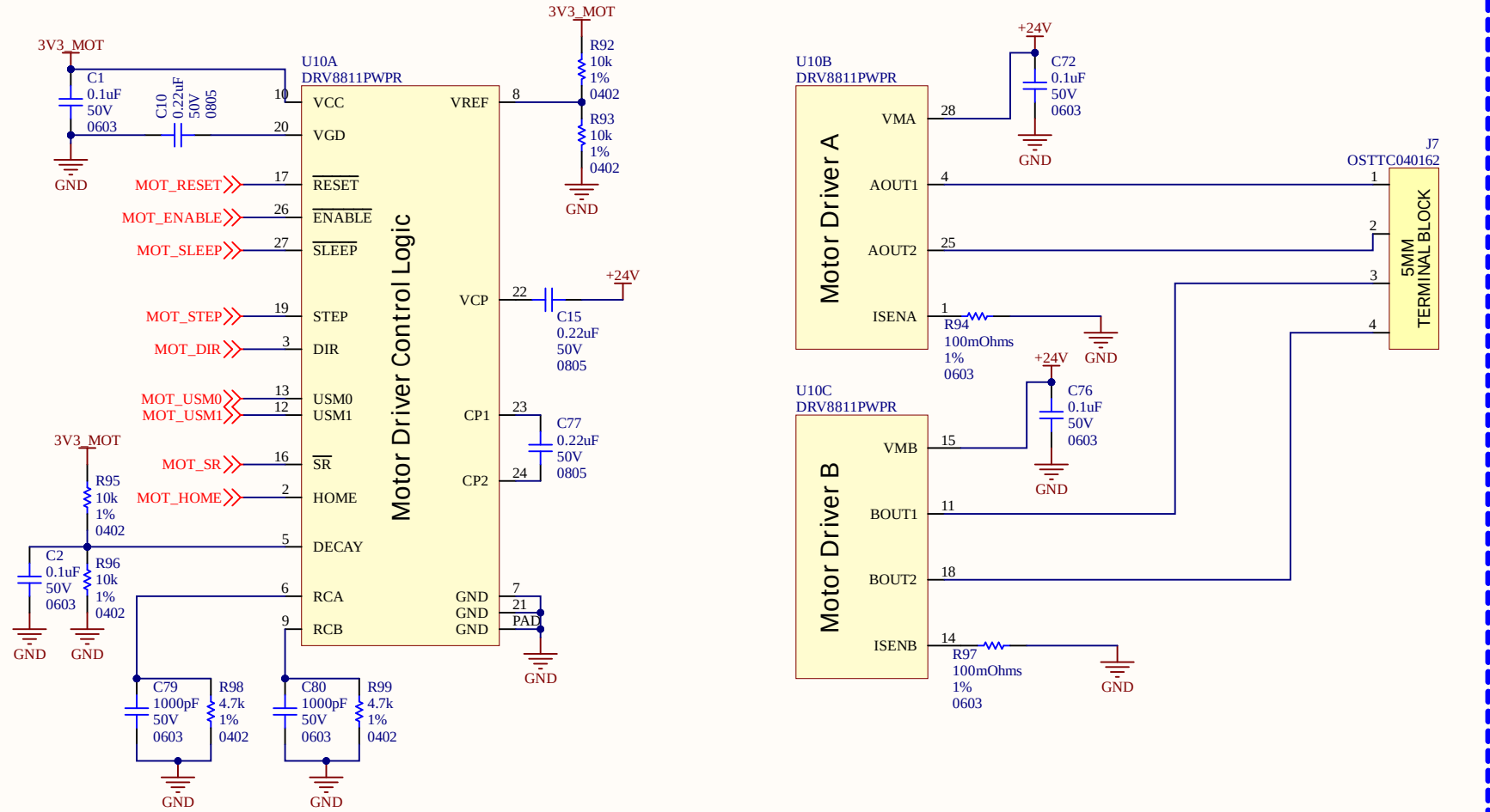
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Global Project			Pleiades		
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DECOUPLAGE



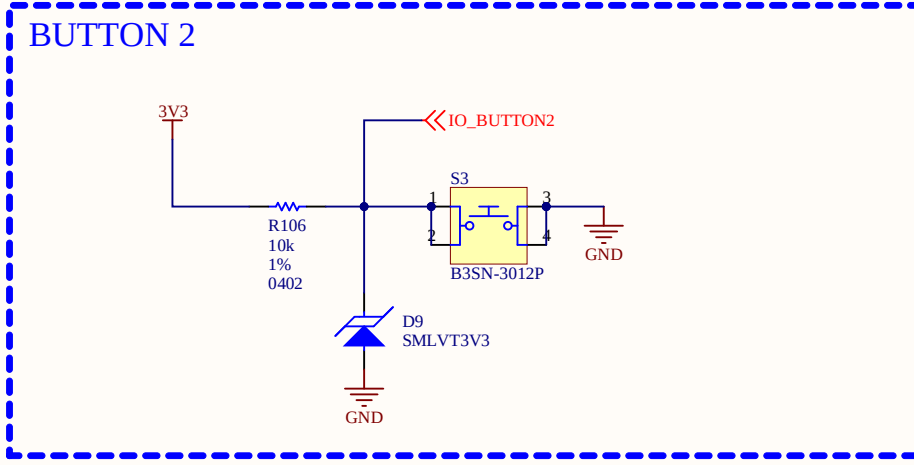
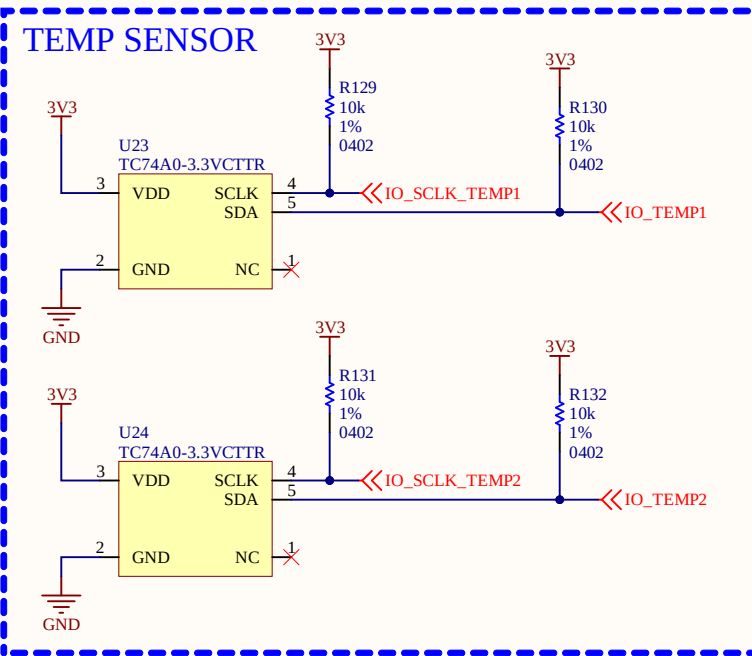
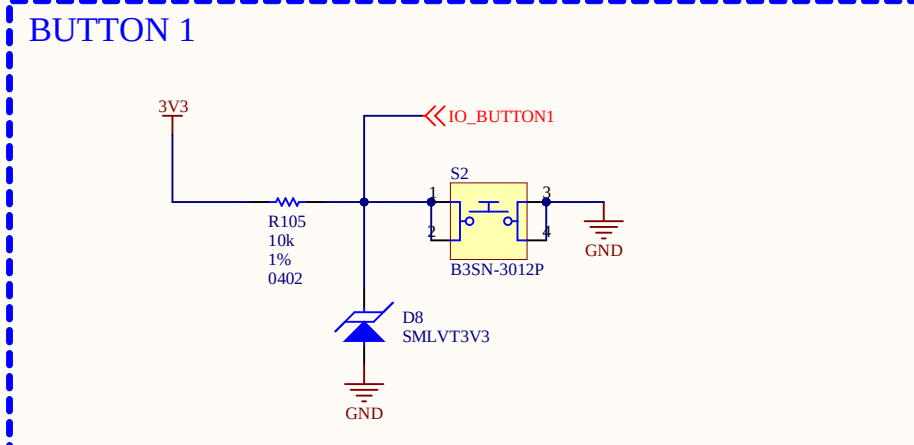
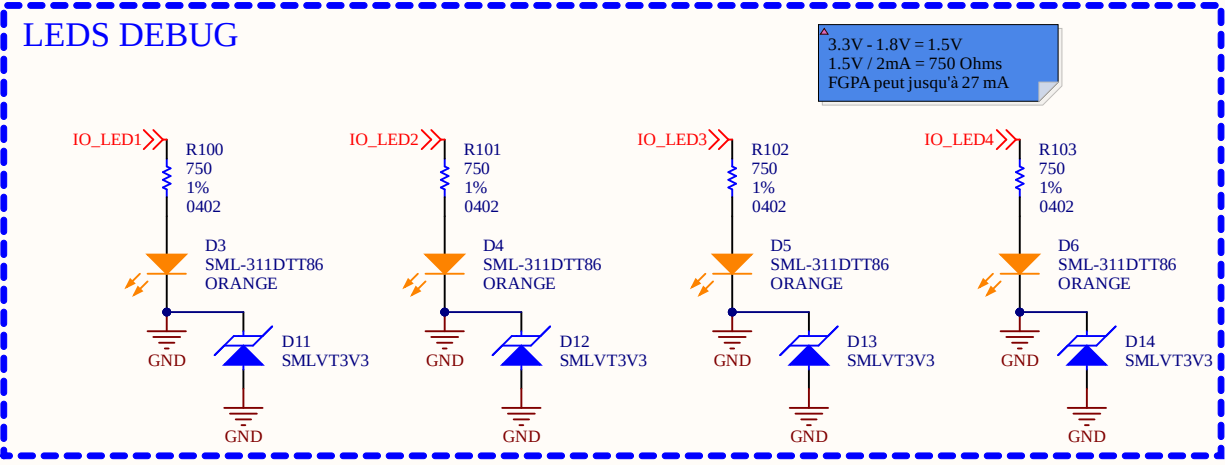
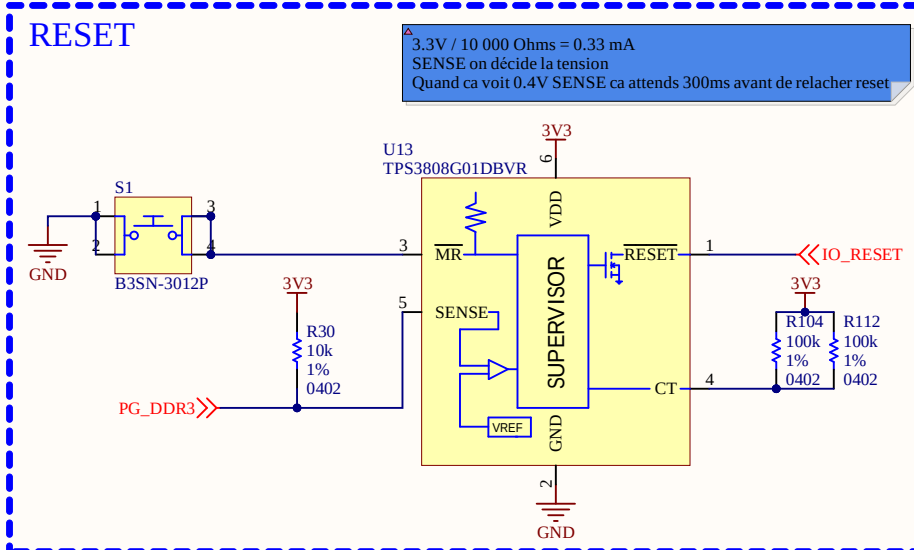
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Global Project			Pleiades		
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S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

MOTOR



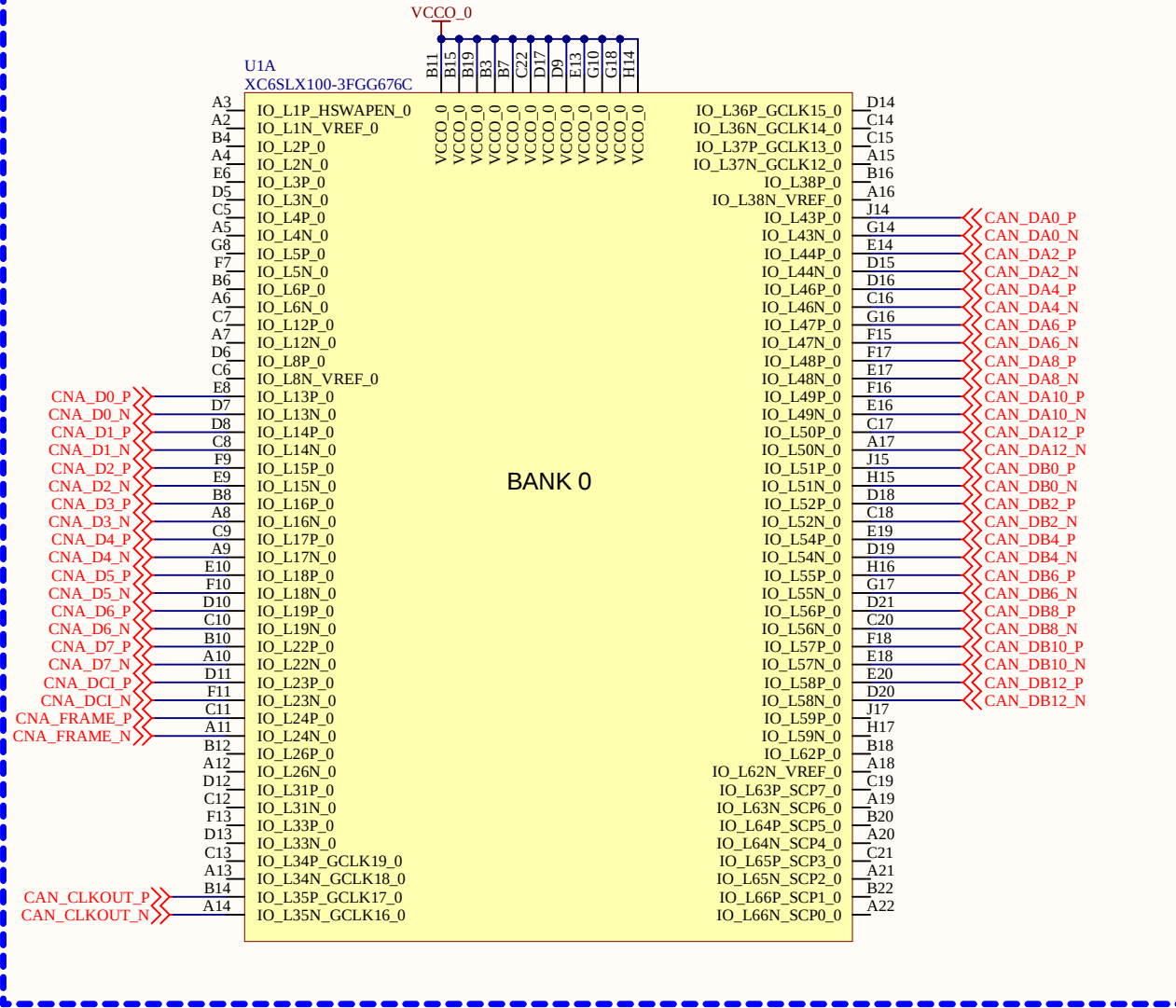
-Tous sur la BANK1
-ISENS = 1.65/(8x1,9A) = 0,108 Ohms

Sheet Name		MOTORS	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
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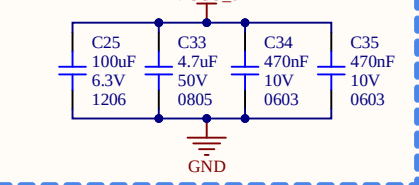


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Global Project			Pleiades		
Size	Group		Revision		
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Filename			Designers		
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FPGA

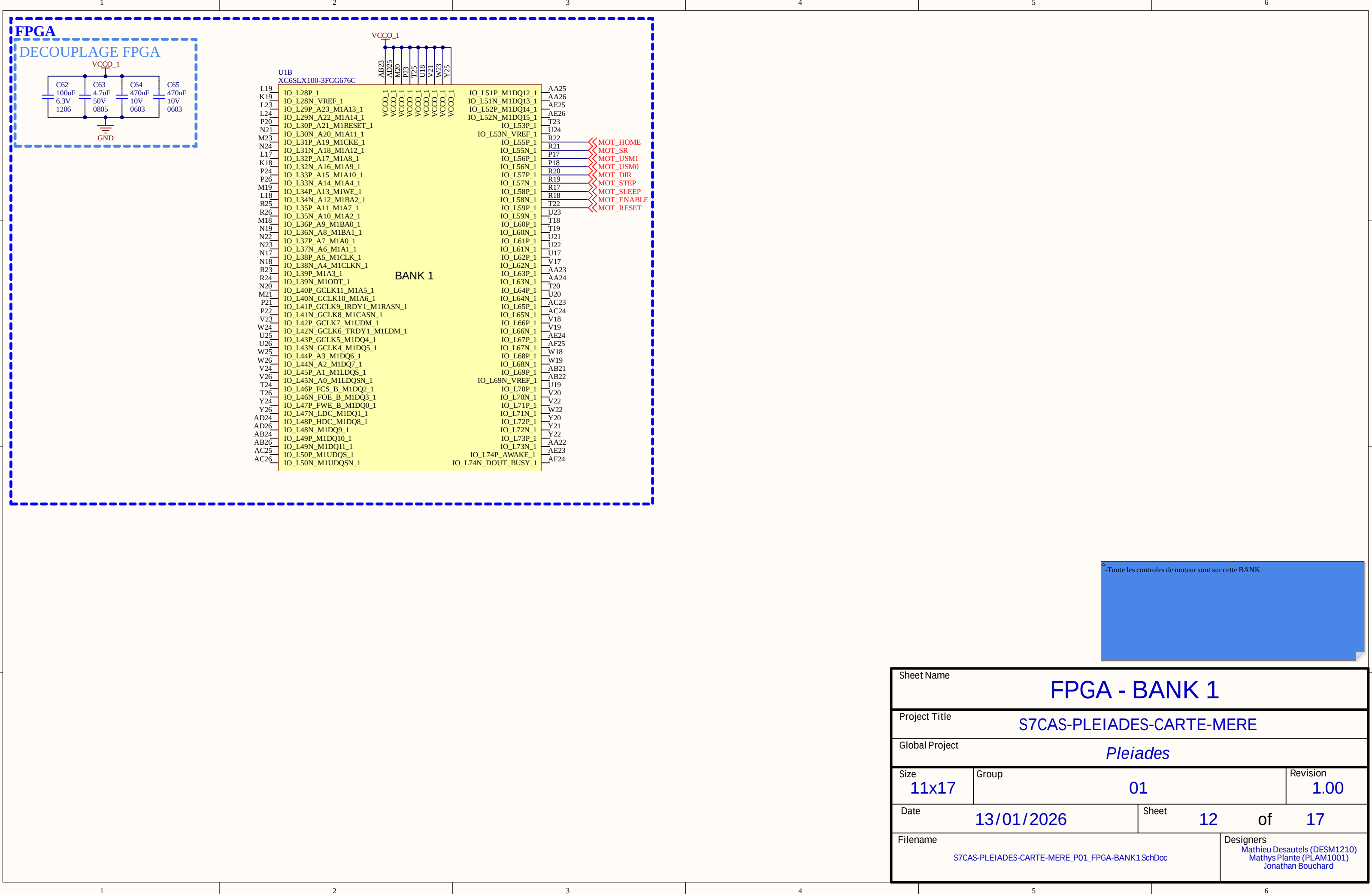


DECOUPLAGE FPGA

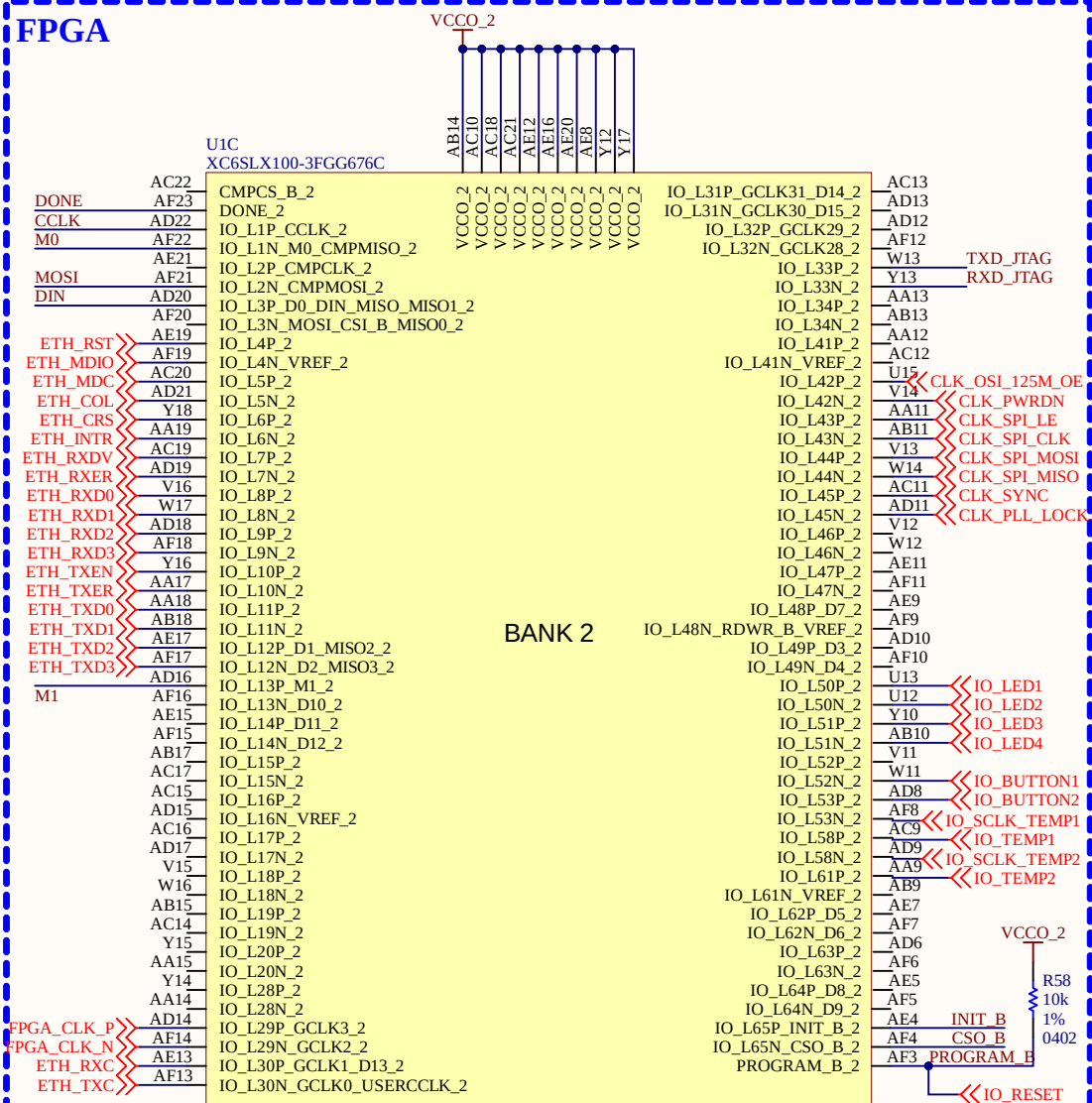


-LVDS seulement sur BANK0 et BANK2
-Toute les CAN et CNA sont mit sur cette BANK

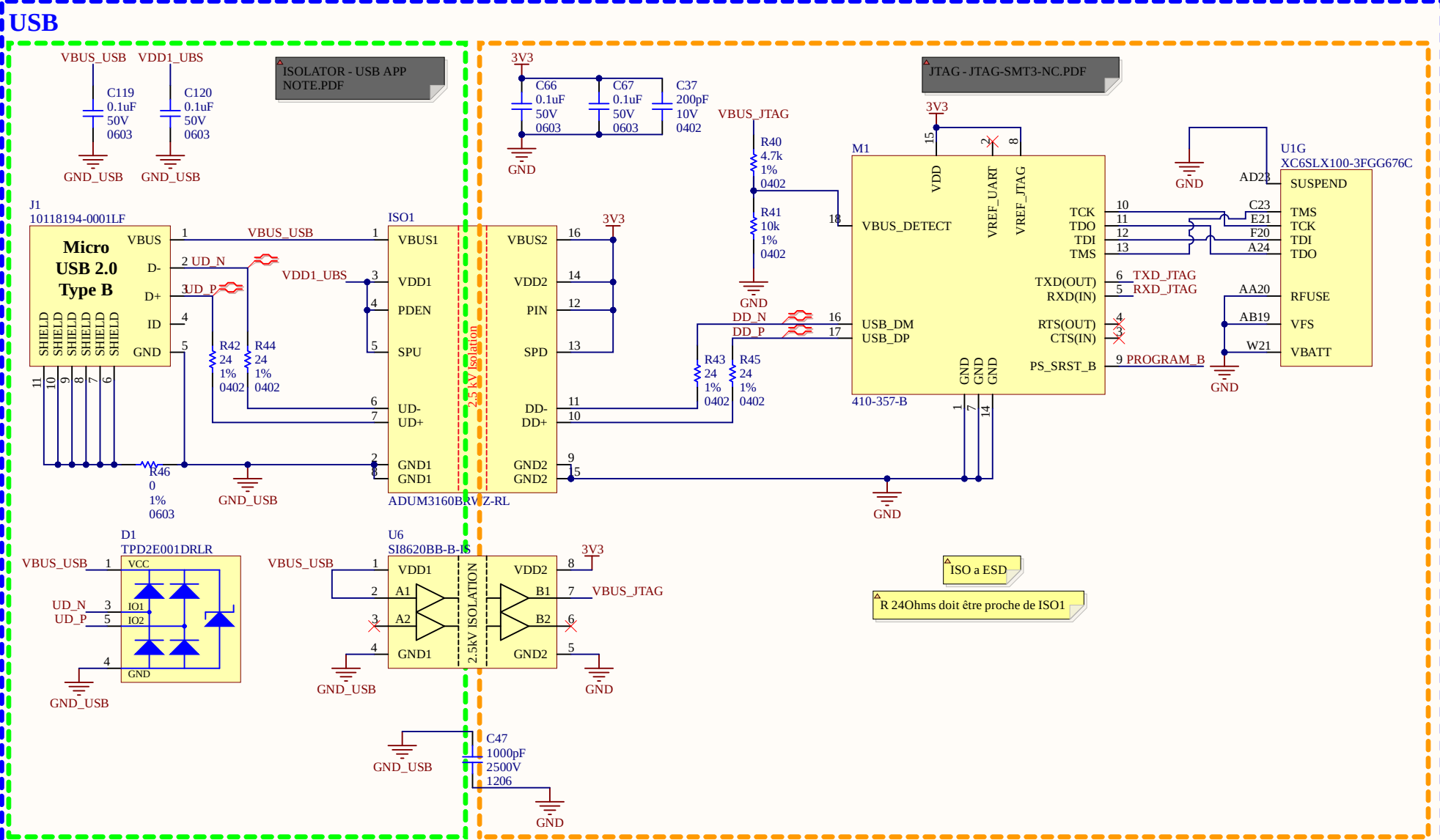
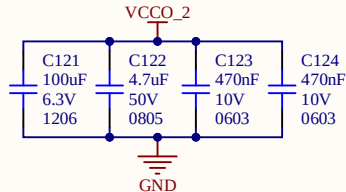
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Size	Group			Revision	
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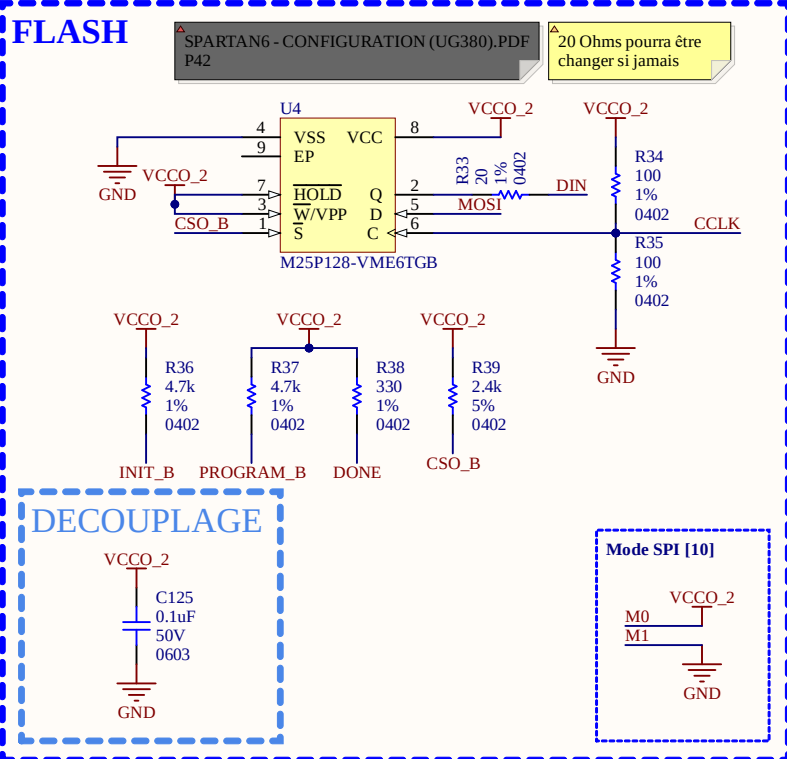
FPGA



DECOUPLAGE FPGA



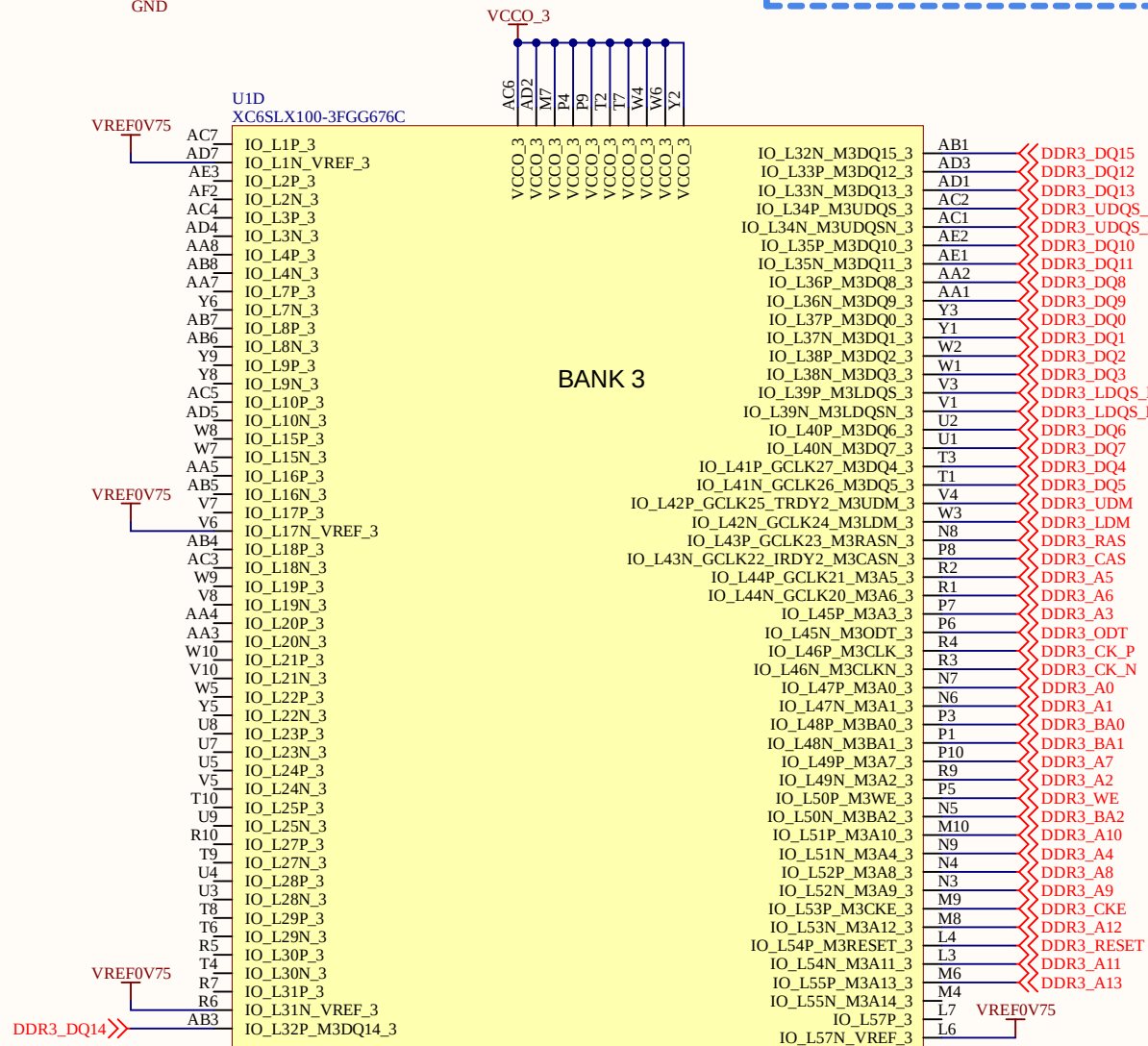
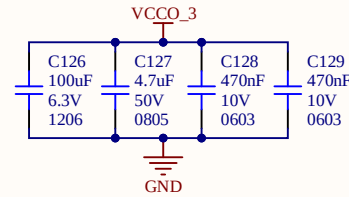
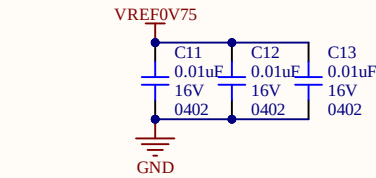
FLASH



- Ajout de ESD pour la protection
 - Protection isolé pour le VBUS
 - Condensateur 2.5kV pour isoler les GND (même force que les isolateurs)
 - Paires différentiel sur le D+ et D-, avec des résistance de 240ohms (comme l'exemple)
 - Isolation du côté USB au lieu d'être du côté board (plus petit comme chip à comparer du ISO de 6 fils)
 - If AES encryption is unused, VBATT can be tied to either, VCCAUX or ground, or left unconnected.
 - VFS and RFUSE are tied to ground because we are not using Efuses
-
- M0 et M1 choisi pour avoir le mode SPI
 - Ajout d'une résistance de 20 ohms sur Q
 - Toute la logique de code est sur la BANK ainsi que toute la programmation

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

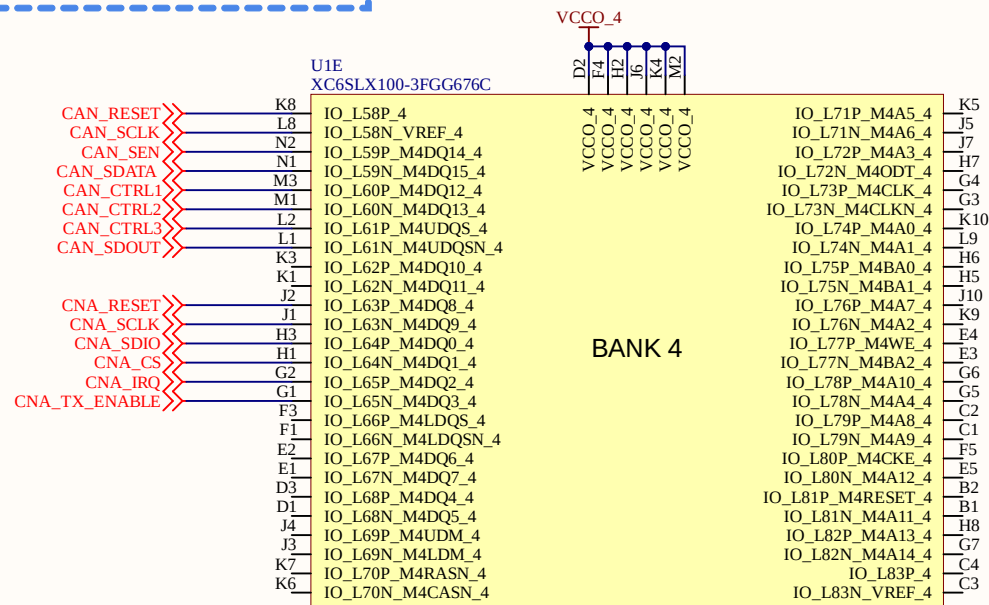
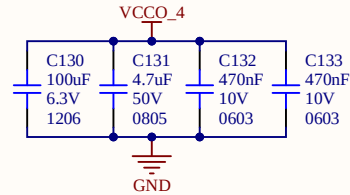


△ -Toute la DDR3 est sur cette BANK

Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

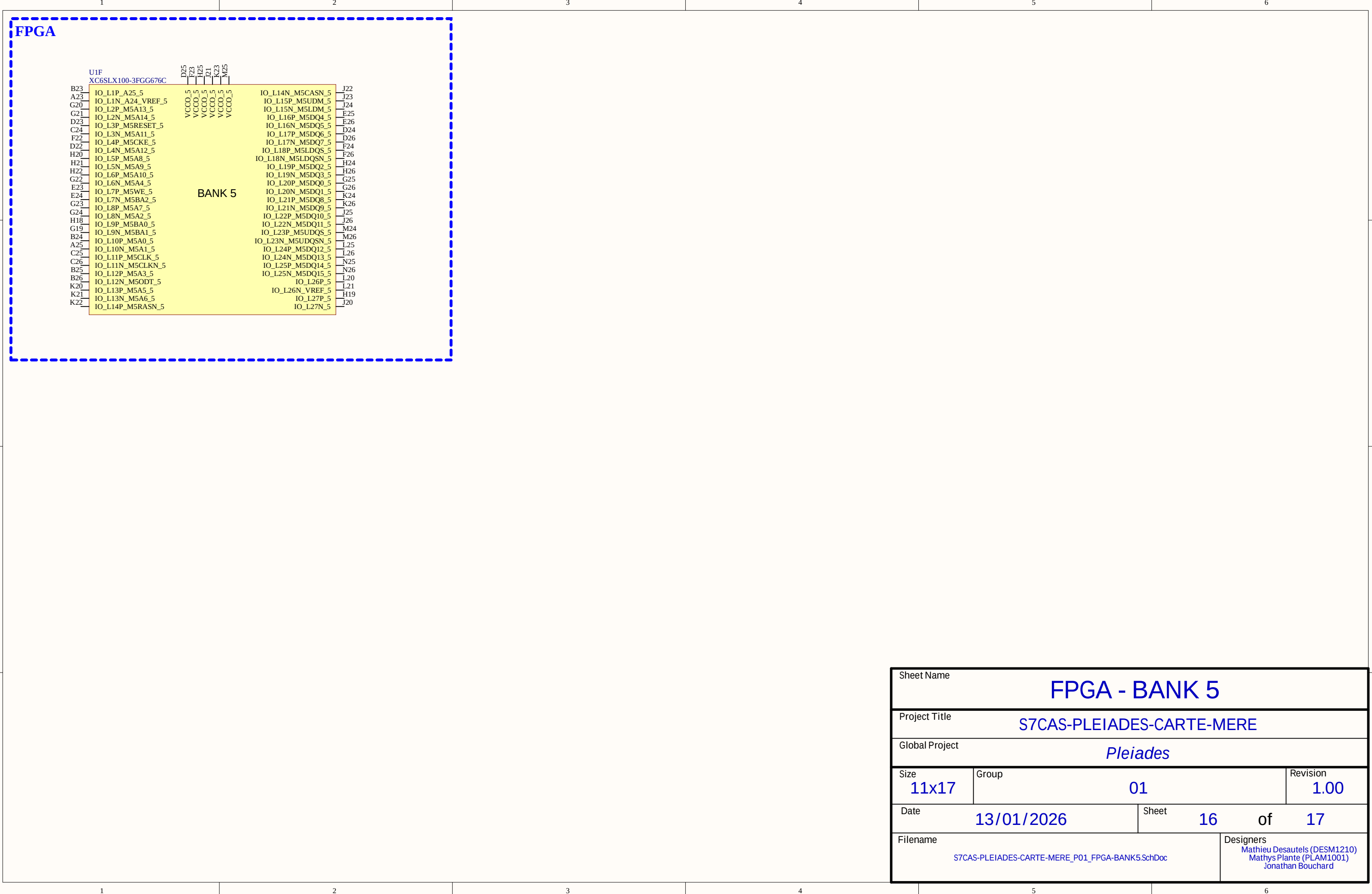
FPGA

DECOUPLAGE FPGA



- Toute le controle du CNA et CAN sont sur cette BANK
- Leur tension n'est pas la même que celle de leur données

Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



A

B

C

D

A

B

C

D

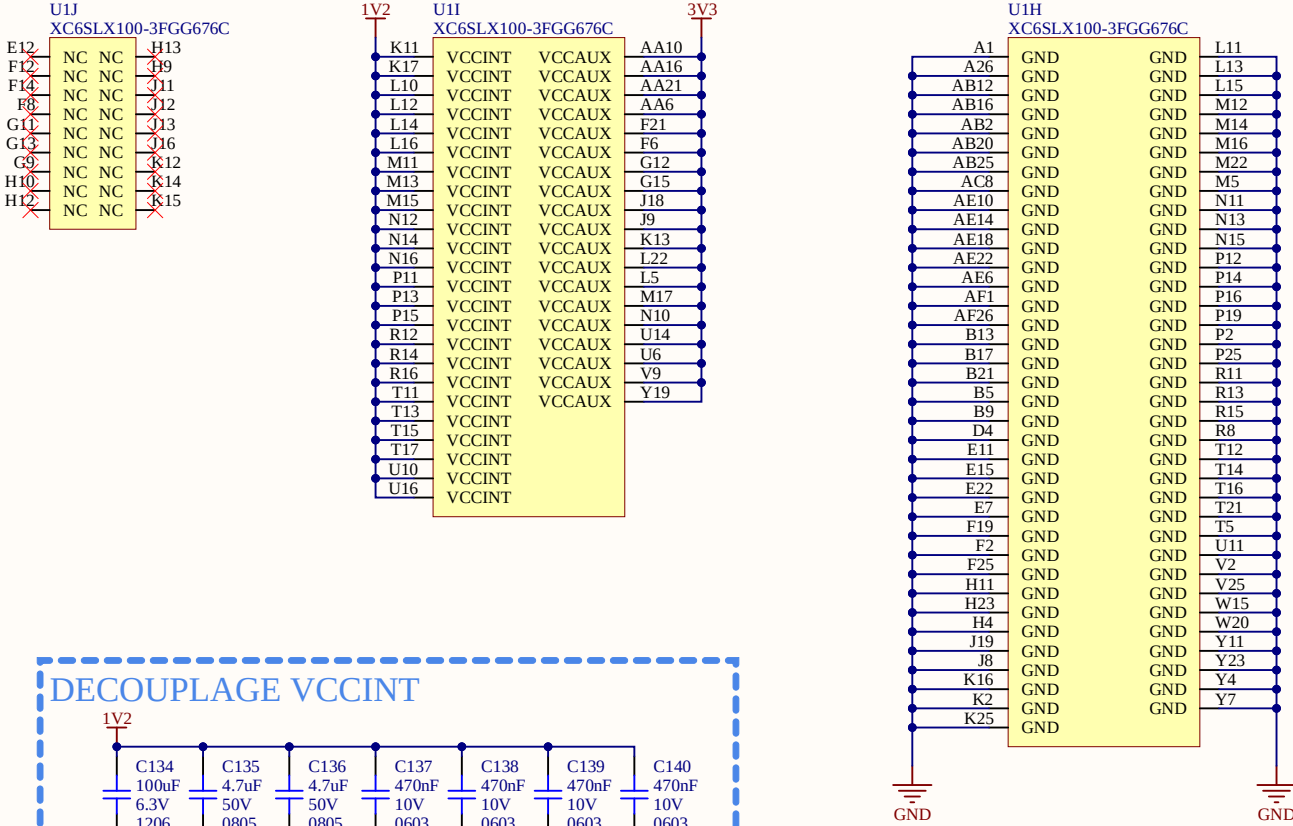


Table 2-1: Required PCB Capacitor Quantities per Device ⁽¹⁾⁽³⁾ (Continued)												
Package	Device (XCSS)	V _{CCINT} In µF	V _{CCAUX} In µF	V _{CC0} Bank 0 In µF	V _{CC0} Bank 1 In µF	V _{CC0} Bank 2 In µF	V _{CC0} Bank 3 In µF	V _{CC0} Bank 4 In µF	V _{CC0} Bank 5 In µF	Total (2)		
		100	4.7	0.47	100	4.7	0.47	100	4.7	0.47		
FG(G)484: LX75		1	2	3	1	2	4	1	1	3	33	
FG(G)484: LX75T		1	2	3	1	2	4	1	1	3	32	
FG(G)484: LX100		1	2	4	1	1	2	1	1	3	33	
FG(G)484: LX100T		1	2	4	1	1	2	1	1	3	33	
FG(G)484: LX150		2	3	6	1	2	3	1	1	3	37	
FG(G)484: LX150T		2	3	6	1	2	3	1	1	3	37	
FG(G)676: LX45		1	1	2	1	2	5	1	1	3	33	
FG(G)676: LX75		1	2	3	2	3	6	1	1	3	45	
FG(G)676: LX75T		1	2	3	1	2	5	1	1	3	40	
FG(G)676: LX100		1	2	4	2	3	6	1	1	3	46	
FG(G)676: LX100T		1	2	4	1	2	5	1	1	3	41	
FG(G)676: LX150		2	3	6	2	3	6	1	1	3	50	
FG(G)676: LX150T		2	3	6	1	2	5	1	1	3	45	
FG(G)900: LX100T		1	2	4	2	3	6	1	1	3	47	
FG(G)900: LX150		2	3	6	2	4	7	1	1	5	57	
FG(G)900: LX150T		2	3	6	2	3	7	1	1	4	54	

Sheet Name			FPGA - ALIM		
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